Date:

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Latihan IV

Bagian A

1. Sederhanakan bilangan imajiner berikut !

a. $j^3 = j^2 j = -1\sqrt{-1}$

b. $j^4 = (j^2)^2 = (-1)^2 = 1$

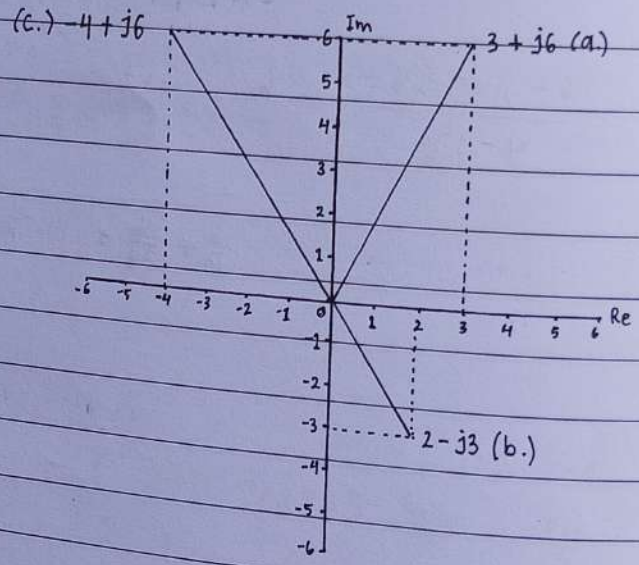
c. $j^5 = (j^2)^2 j = (-1)^2\sqrt{-1} = \sqrt{-1}$

2. Tunjukkan bilangan kompleks berikut dalam diagram Argand !

a. $3 + j6$

b. $2 - j3$

c. $-4 + j6$



3. Tulislah bentuk sekawan (conyugate) dari lawan bil. kompleks berikut.

a. $4 + j = 4 - j$

b. $-4 - j = -4 + j$

c. $3 - j2 = 3 + j2$

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4. Nyatakan bilangan kompleks berikut dalam bentuk polar.

a. $-1 + j$

$z = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$

$\theta = \tan^{-1}\left(\frac{1}{-1}\right)$

$= \tan^{-1} -1$

$\theta = -45°$

Jadi, bentuk polarnya $z = \sqrt{2}\,(\cos 45° - j \sin 45°)$

$z = \sqrt{2} \angle -45°$

b. $-1 + j$

$z = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$

$\theta = \tan^{-1}\left(\frac{1}{-1}\right)$

$= \tan^{-1} -1$

$\theta = -45°$

Jadi, bentuk polarnya $z = \sqrt{2}\,(\cos 45° - j \sin 45°)$

$z = \sqrt{2} \angle -45°$

c. $-3 - j2$

$z = \sqrt{(-3)^2 + (-2)^2} = \sqrt{13}$

$\theta = \tan^{-1}\left(\frac{-2}{-3}\right)$

$= \tan^{-1}\left(\frac{2}{3}\right)$

$\theta = 33{,}69°$

Jadi, bentuk polarnya $z = \sqrt{13}\,(\cos 33{,}69° + j \sin 33{,}69°)$

$z = \sqrt{13} \angle 33{,}69°$

5. Nyatakan dalam bentuk Cartesian / rectangular

a. $32\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)$

$r = 32,\ \theta = \frac{3\pi}{4}$

$a = r \cos\theta \rightarrow 32 \cos \frac{3\pi}{4} \rightarrow a = 31{,}97° \ /\ 0{,}17\pi$

$b = r \sin\theta \rightarrow 32 \sin \frac{3\pi}{4} \rightarrow b = 1{,}31° \ /\ 0{,}007\pi$

$z = 31{,}97° + j1{,}31° \ /\ z = 0{,}17\pi + j\,0{,}007\pi$

b. $4 \angle \frac{5\pi}{3}$

$r = 4,\ \theta = \frac{5\pi}{3}$

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$a = r \cos\theta \rightarrow 4 \cos \frac{5\pi}{3} \rightarrow a = 4{,}97^\circ \;/\; 0{,}02\pi$

$b = r \sin\theta \rightarrow 4 \sin \frac{5\pi}{3} \rightarrow b = 0{,}36^\circ \;/\; 0{,}002\pi$

$z = 4{,}97^\circ + j0{,}36^\circ \;/\; z = 0{,}02\pi + j0{,}002\pi$

c. $8 \angle 110^\circ$

$r = 8, \quad \theta = 110^\circ$

$a = r \cos\theta \rightarrow 8 \cos 110^\circ \rightarrow a = -2{,}73^\circ$

$b = r \sin\theta \rightarrow 8 \sin 110^\circ \rightarrow b = 7{,}51^\circ$

$z = -2{,}73^\circ + j\,7{,}51^\circ$

Bagian B

1. Operasikan perkalian berikut dalam bentuk Cartesian

a. $(6-j)(j5) = 6j5 - j^2 5 = j30 + 5$

b. $(2-j)(3+j) = 6 + j2 - j3 - j^2 = 6 - j + 1 = 7 - j$

c. $(-2+j3)(-1-j2) = 2 + j2 - j3 - j^2 6 = 2 - j - j^2 6 = 2 - j + 6 = 8 - j$

2. Operasikan perkalian berikut dalam bentuk Cartesian

a. $\frac{3+j4}{1-j2} = \frac{z_1}{z_2}$

$\frac{z_1}{z_2} \times \frac{\bar{z}_2}{\bar{z}_2} = \frac{3+j4}{1-j2} \times \frac{1+j2}{1+j2}$

$= \frac{3 + j6 + j4 + j^2 8}{1 + j2 - j2 - j^2 4} = \frac{-5 + j10}{5} = \frac{-5}{5} + \frac{j10}{5} = -1 + j2$

b. $\frac{1}{2+j3} + \frac{1}{2-j3}$

$\frac{z_1}{z_2} = \frac{1}{2+j3}$

$\frac{z_1}{z_1} \times \frac{\bar{z}_2}{\bar{z}_2} = \frac{1}{2+j3} \times \frac{2-j3}{2-j3}$

$= \frac{2-j3}{4 - j6 + j6 - j^2 9} = \frac{2-j3}{4+9} = \frac{2-j3}{13} = \ldots$

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$\frac{z_1}{z_2} = \frac{1}{2-j3}$

$\frac{z_1}{z_2} \times \frac{\bar{z}_2}{\bar{z}_2} = \frac{1}{2-j3} \times \frac{2+j3}{2+j3}$

$= \frac{2+j3}{4+j6-j6-j^2 9} = \frac{2+j3}{4+9} = \frac{2+j3}{13}$

$\frac{2-j3}{13} + \frac{2+j3}{13} = \frac{4}{13}$

c. $\frac{1}{3+j} - \frac{1}{1+j7}$

$\frac{z_1}{z_2} = \frac{1}{3+j}$

$\frac{z_1}{z_2} \times \frac{\bar{z}_2}{\bar{z}_2} = \frac{1}{3+j} \times \frac{3-j}{3-j}$

$= \frac{3-j}{9-j3+j3.-j^2} = \frac{3-j}{9+1} , \frac{3-j}{10}$

$\frac{z_1}{z_2} = \frac{1}{1+j7}$

$\frac{z_1}{z_2} \times \frac{z_2}{z_2} = \frac{1}{1+j7} \times \frac{1-j7}{1-j7}$

$= \frac{1-j7}{1-j7+j7-j^2 49} = \frac{1-j7}{1+49} = \frac{1-j7}{50}$

$\frac{3-j}{10} - \frac{1-j7}{50} \rightarrow \frac{15-j5}{50} - \frac{1-j7}{50} = \frac{14+j2}{50} = \frac{7+j}{25} = \frac{7}{25} + \frac{j}{25}$

3. Selesaikan :

a. $(12+j) - (4+j5)$

$z_1 - z_2 = (12-4) + j(1-5) = 8-j4$

b. $(3+j2)(4-j3) = 12 - j9 + j8 - j^2 6$

$= 12 - j + 6 = 18 - j$

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c. $(2-j5)^2 = (2-j5)(2-j5)$

$= 4 - j10 - j10 + j^2 25 = 4 - j20 - 25 = -21 - j20$

4. Diketahui $z_1 = 3 + j2$, $z_2 = 2 - j3$, dan $z_3 = -4 + j$

Hitung :

a. $z_1 z_2 = (3+j2)(2-j3)$

$= 6 - j9 + j4 - j^2 6 = 6 - j5 + 6 = 12 - j5$

b. $z_2 z_1 z_3 = (2-j3)(3+j2)(-4+j)$

$z_2 z_1 = 6 + j4 - j9 - j^2 6 = 6 - j5 + 6 = 12 - j5$

$z_2 z_1 z_3 = (12-j5)(-4+j)$

$= -48 + j12 + j20 - j^2 5 = -48 + j32 + 5 = -43 + j32$

c. $\frac{z_3}{z_1} = \frac{-4+j}{3+j2}$

$\frac{z_3}{z_1} = \frac{-4+j}{3+j2} \times \frac{3-j2}{3-j2}$

$= \frac{-12 + j8 + j3 - j^2 2}{9 - j6 + j6 - j^2 4} = \frac{-12 + j11 + 2}{9 + 4} = \frac{10 + j11}{13} = \frac{10}{13} + \frac{j11}{13}$